

HPC Pilot Job Checklist

Project: prostate_mri_task_iqa. Use this when you return to bash after submitting or monitoring a GPU job on the UCL CS cluster.

Quick rule

If you are watching logs with `tail -f` and the terminal seems stuck, press `Ctrl+C`. That only stops watching the log; it does not cancel the cluster job.

Checklist commands

1. Go to the project folder

```
cd /home/yulong/prostate_mri_task_iqa
```

2. Reactivate the cluster Python and project environment

```
source /share/apps/source_files/python/YOUR_REAL_PYTHON_SOURCE_FILE
source /home/yulong/prostate_mri_task_iqa/.venv/bin/activate
export PYTHONPATH=/home/yulong/prostate_mri_task_iqa/src
```

3. Check whether your pilot job is still running

```
qstat -u "$USER"
```

4. If the job is running, inspect recent logs

```
tail -n 40 logs/pilot_pirads.out
tail -n 40 logs/pilot_pirads.err
```

5. Optional live monitoring

```
tail -f logs/pilot_pirads.out
```

6. If qstat shows nothing, inspect finished job outputs

```
cat logs/pilot_pirads.err
tail -n 80 logs/pilot_pirads.out
ls -lh runs/pilot_promis_pirads
```

7. Check Grid Engine accounting for the pilot job

```
qacct -j 6983787 | grep -E "failed|exit_status|maxvmem|ru_wallclock"
```

How to interpret the results

What you see	Meaning	What to do
qstat shows pilot_pira with state <code>r</code>	The job is running on a GPU node.	Keep waiting or inspect logs with <code>tail -n</code> .
qstat shows nothing	The job finished, failed, or left the queue.	Check logs, output files, and qacct.
qacct failed=0 and exit_status=0	The scheduler considers the job successful.	Inspect the run folder and metrics.
logs/pilot_pirads.err contains Python/source errors	The environment was not loaded correctly.	Fix the source line in the job script and resubmit.

Expected successful files

runs/pilot_promis_pirads/best.pt
runs/pilot_promis_pirads/last.pt
runs/pilot_promis_pirads/val_predictions.csv

Next step after a successful pilot

If best.pt, last.pt, and val_predictions.csv exist, the pilot training path works. The next workflow stage is to submit full downstream or out-of-fold jobs, then generate task-derived IQA labels and train the binary or ternary IQA model. Keep the locked test split untouched until final evaluation.

Important placeholders

Replace YOUR_REAL_PYTHON_SOURCE_FILE with the exact Python source file provided by the cluster. Do not leave the placeholder in a qsub script.

The accounting command uses the current pilot job ID 6983787. For a new job, replace that number with the new ID printed by qsub.